<u>Step 3</u>: If δ is suff. small, then for $y \in V$ we have $u^{-1}(u(y)) = \{y\}$.

<u>Pf</u>: If δ suff. small, then for all $r > 0$ s.t. $B_r(y) \subset B_R(p)$, u is a $\frac{r}{2}$-GH approximation $B_r(y) \longrightarrow B_r(0)$

i.e. $d(u(\partial B_r(y)), u(y)) \geq \frac{r}{2} > 0$

<u>Step 4</u>: u on $B_R(p)$ is a Ψ-GH approx

so u maps pair $(B_R(p), B_R(p) \setminus B_{(1-\Psi)R}(p))$

to $(B_{R+\Psi}(0), B_{R+\Psi}(0) \setminus B_{(1-2\Psi)R}(0))$

mod 2 degree $= 1 \implies u$ map onto $B_{(1-2\Psi)R}(0)$

$$(1-\Psi)\,\mathrm{Vol}(B_R(p)) \leq \mathrm{Vol}\left(u(B_R(p))\right) \leq \int_{B_R(p)} |\det Du|$$

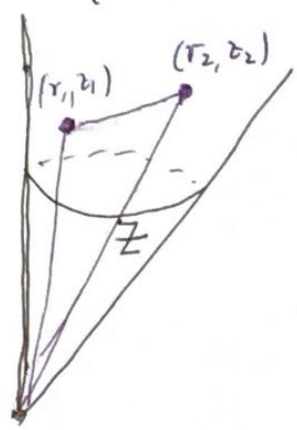
$$= \mathrm{Vol}(B_R(p)) \fint_{B_R(p)} |\det Du| \leq \mathrm{Vol}(B_R(p))(1+\Psi)$$

$\square$

Almost volume cones

For metric space Z, $C(Z)$ = metric cone: completion of $(0,\infty)\times Z$ with metric

$$d\big((r_1,z_1),(r_2,z_2)\big) = \begin{cases} r_1^2+r_2^2-2r_1r_2\cos d(z_1,z_2) & \text{if } d(z_1,z_2)\le\pi \\ r_1+r_2 & \text{if } d(z_1,z_2)>\pi \end{cases}$$



Write z^* for vertex of $C(Z)$ and $C_{0,r}(Z) = B_r(z^*) \subset C(Z)$

<u>Theorem</u> Assume $\mathrm{Ric}_{M^n} \geq -(n-1)\delta$, $\partial B_{2R}(p)\neq\emptyset$ and

$$(1-\delta)\frac{\mathrm{Vol}\,B_R(p)}{\mathrm{Vol}\,B_R(\underline{0})} \le \frac{\mathrm{Vol}\,\partial B_R(p)}{\mathrm{Vol}\,\partial B_R(\underline{0})}, \quad \underline{0}\in M^n_{-\delta}$$

Then $\exists$ length space Z, $\mathrm{diam}(Z) < \pi + \Psi$ s.t.

$$d_{GH}\big(B_R(p),\, C_{0,R}(Z)\big) < \Psi$$

for $\Psi = \Psi(\delta \mid n, R)$.

<u>Proof sketch</u> Similar to almost splitting theorem.

Let $f : B_R(p) \to \mathbb{R}$ be defined by

$$\Delta f = 1 \quad \text{on } B_R(p)$$

$$f = \underline{u}(R) \quad \text{on } \partial B_R(p)$$

$$\left(\begin{array}{l} \underline{u} \text{ is fn. on } M^n_{-\delta} \text{ with } \underline{\Delta}\,\underline{u} = 1 \\ \text{on } \mathbb{R}^n \quad \underline{u} = \frac{r^2}{2n} \end{array} \right)$$

$$\int_{B_R(p)} \Delta f = \operatorname{Vol} B_R(p).$$

$$\int_{B_R(p)} \Delta \underline{u} = \int_{\partial B_R(p)} \nabla_N \underline{u} = \int_{\partial B_R(p)} \underline{u}'(R)$$

$$= \frac{\operatorname{Vol}(\partial B_R(p))}{\operatorname{Vol}(\partial B_R(\underline{o}))} \int_{\partial B_R(\underline{o})} \underline{u}'(R)$$

$$= \frac{\operatorname{Vol}(\partial B_R(p))}{\operatorname{Vol}(\partial B_R(\underline{o}))} \int_{B_R(\underline{o})} \underline{\Delta}\,\underline{u} = \frac{\operatorname{Vol}(\partial B_R(p))}{\operatorname{Vol}(\partial B_R(\underline{o}))} \operatorname{Vol} B_R(\underline{o})$$

$$\geq (1-\delta) \operatorname{Vol} B_R(p)$$

$$\Rightarrow \quad 0 \leq \int_{B_R(p)} \Delta(f - \underline{u}) \leq \delta$$

$$\Delta \underline{u} \leq \underbrace{\underline{\Delta}\,\underline{u}}_{=\Delta f} = 1 \quad \text{by Laplacian comp.}$$

$\rightsquigarrow$ $|\underline{u} - f| \leq \Psi$

(using max principle, Poincaré ineq., grad. estimate) on $B_{(1-\Psi)R}(p)$.

<u>Note</u>: $\underline{u} = \frac{r^2}{2n} \pm \Psi$

$|\nabla \underline{u}| = \frac{r}{n} \pm \Psi$

$$\Longrightarrow \quad \fint_{B_R(p)} \left| |\nabla f|^2 - \frac{2f}{n} \right| \leq \Psi$$

Bochner formula with cutoff fn. ϕ:

$$\Psi \geq \fint_{B_R(p)} \frac{1}{2} \Delta \phi \left(|\nabla f|^2 - \frac{2f}{n} \right) = \fint_{B_R(p)} \frac{1}{2} \phi \, \Delta \left(|\nabla f|^2 - \frac{2f}{n} \right)$$

$$\geq \fint_{B_R(p)} \phi \left(|\nabla^2 f|^2 - (n-1) \delta |\nabla f|^2 - \frac{1}{n} \right)$$

$$\Longrightarrow \quad \int_{B_{(1-\Psi)R}(p)} \underbrace{\left| \nabla^2 f - \frac{1}{n} g \right|^2}_{= |\nabla^2 f|^2 - \frac{2}{n} \Delta f + \frac{1}{n}} \leq \Psi$$

Rest of proof is use segment inequality to convert this into distance estimates. $\square$

<u>Note</u> By relative volume comparison if

$$(1-\delta)\frac{\mathrm{Vol}(B_{\frac{1}{2}R}(p))}{\mathrm{Vol}(B_{\frac{1}{2}R}(\underline{0}))} \leq \frac{\mathrm{Vol}(B_R(p))}{\mathrm{Vol}(B_R(\underline{0}))} \quad (*)$$

then $\forall \eta \leq \frac{1}{2}$ we have

$$\left(1-\Psi(\delta|\eta,n,R)\right)\frac{\mathrm{Vol}(B_{R'}(p))}{\mathrm{Vol}(B_{R'}(\underline{0}))} \leq \frac{\mathrm{Vol}(\partial B_{R'}(p))}{\mathrm{Vol}(\partial B_{R'}(\underline{0}))}$$

for $R' = (1-\eta)R$

$\Rightarrow$ conclusion of Thm holds under $(*)$.

<u>Theorem</u> Let $\mathrm{Ric}_{M^n} \geq -(n-1)\delta$

$\mathrm{Vol}(B_R(p)) \geq (1-\delta)\mathrm{Vol}(B_R(0))$ $\quad 0 \in \mathbb{R}^n$

$\Rightarrow$ $d_{GH}(B_R(p), B_R(0)) < \Psi(\delta|n,R)$